

## DATA CLEANING

```
import pandas as pd

# Load the dataset
df = pd.read_csv("startups-expansions-modified (1).csv")

# 1. Check the dataset shape
print("Shape:")
print(df.shape)

# 2. Check column names
print("\nColumns:")
print(df.columns)

# 3. Check data types
print("\nData Types:")
print(df.dtypes)

# 4. Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# 5. Check duplicated rows
print("\nDuplicated Rows:")
print(df.duplicated().sum())
```

# 6. Remove unnecessary column

```
df = df.drop(columns=["Unnamed: 0"])
```

# 7. Convert ROMS% from string to numeric

```
df["ROMS%"] = df["ROMS%"].str.replace("%", "").astype(float)
```

# 8. Check the cleaned dataset

```
print("\nCleaned Data:")
```

```
print(df.head())
```

# 9. Check data types again

```
print("\nData Types After Cleaning:")
```

```
print(df.dtypes)
```

# 10. Final missing values check

```
print("\nMissing Values After Cleaning:")
```

```
print(df.isnull().sum())
```

## Exploratory Data Analysis (EDA )

# 1. Display the first 5 rows

```
print("First 5 Rows:")
```

```
print(df.head())
```

# 2. Display the last 5 rows

```
print("\nLast 5 Rows:")
```

```
print(df.tail())
```

# 3. Check the shape of the dataset

```
print("\nShape:")
```

```
print(df.shape)
```

# 4. Get information about the dataset

```
print("\nDataset Information:")
```

```
print(df.info())
```

# 5. Generate descriptive statistics

```
print("\nDescriptive Statistics:")
```

```
print(df.describe())
```

# 6. Count New vs Old Expansion

```
print("\nNew vs Old:")
```

```
print(df["NEW EXPANSION"].value_counts())
```

# 7. Calculate the proportion of New vs Old

```
print("\nNew vs Old Proportion:")
```

```
print(df["NEW EXPANSION"].value_counts(normalize=True))
```

# 8. Count stores by Sales Region

```
print("\nStores by Region:")
```

```
print(df["SALES REGION"].value_counts())
```

# 9. Count unique cities

```
print("\nNumber of Unique Cities:")
```

```
print(df["CITY"].nunique())
```

# 10. Count unique states

```
print("\nNumber of Unique States:")
```

```
print(df["STATE"].nunique())
```

## ANALYSIS DATA

```
# New vs Old Expansion Analysis
```

```
# 1. Count New and Old stores
```

```
expansion_counts = df["NEW EXPANSION"].value_counts()
```

```
print("Number of Stores:")
```

```
print(expansion_counts)
```

```
# 2. Calculate the percentage of New and Old stores
```

```
expansion_proportions = (
```

```
    df["NEW EXPANSION"].value_counts(normalize=True) * 100
)
```

```
print("\nPercentage of Stores:")
```

```
print(expansion_proportions)
```

```
# 3. Average Revenue by Expansion Type
```

```
avg_revenue = df.groupby("NEW EXPANSION")["REVENUE"].mean()
```

```
print("\nAverage Revenue:")
```

```
print(avg_revenue)
```

# 4. Average Profit by Expansion Type

```
avg_profit = df.groupby("NEW EXPANSION")["PROFIT"].mean()
```

```
print("\nAverage Profit:")
```

```
print(avg_profit)
```

# 5. Average Marketing Spend by Expansion Type

```
avg_marketing = df.groupby("NEW EXPANSION")["MARKETING SPEND"].mean()
```

```
print("\nAverage Marketing Spend:")
```

```
print(avg_marketing)
```

# 6. Average ROMS by Expansion Type

```
avg_roms = df.groupby("NEW EXPANSION")["ROMS"].mean()
```

```
print("\nAverage ROMS:")
```

```
print(avg_roms)
```

# 7. Average ROMS% by Expansion Type

```
avg_roms_percentage = df.groupby("NEW EXPANSION")["ROMS%"].mean()
```

```
print("\nAverage ROMS%:")  
print(avg_roms_percentage)
```

# 8. Create a complete summary table

```
expansion_summary = df.groupby("NEW EXPANSION").agg({  
    "REVENUE": ["mean", "sum"],  
    "PROFIT": ["mean", "sum"],  
    "MARKETING SPEND": ["mean", "sum"],  
    "ROMS": "mean",  
    "ROMS%": "mean"  
})
```

```
print("\nExpansion Summary:")  
print(expansion_summary)
```